

Fig 200: FANNO-Dev Complete Analysis Dashboard (200 Figures)

Dataset Overview	
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Total Samples	153,351
Single-turn	134,643
Multi-turn	18,708
Pipelines	8
Unique Domains	2,297
Question Types	25
Est. Cost	\$2,245
Rejection Rate	27.4%

Diversity Metrics	
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Vendi Score	182.75
Avg Cos Dist	0.951
Cross-Source	0.961
Intrinsic Dim	50+
K-Means Clusters	20
NMF Topics	20
Entities	52,525
Bloom Types	6

Pipeline Highlights	
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Self-Inversio	2,288 domains, hull=0.537
Document QA	L00=+12.3, 76.6% exc ent
Complex QA	5,187/1K refs, bloom=3.27
Creative	bloom=4.25, 84% prose
Reasoning	coherence=0.413
Code QA	7.25 struct features
Math QA	grade=7.3, density=42.5

Deliverables	
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Figures (PNG+PDF)	200
LaTeX Tables	27
JSON Reports	80+
Paper Draft	1
Dataset Card	1
Training Recipe	1
Git Commits	175+
Analysis Sessions	8

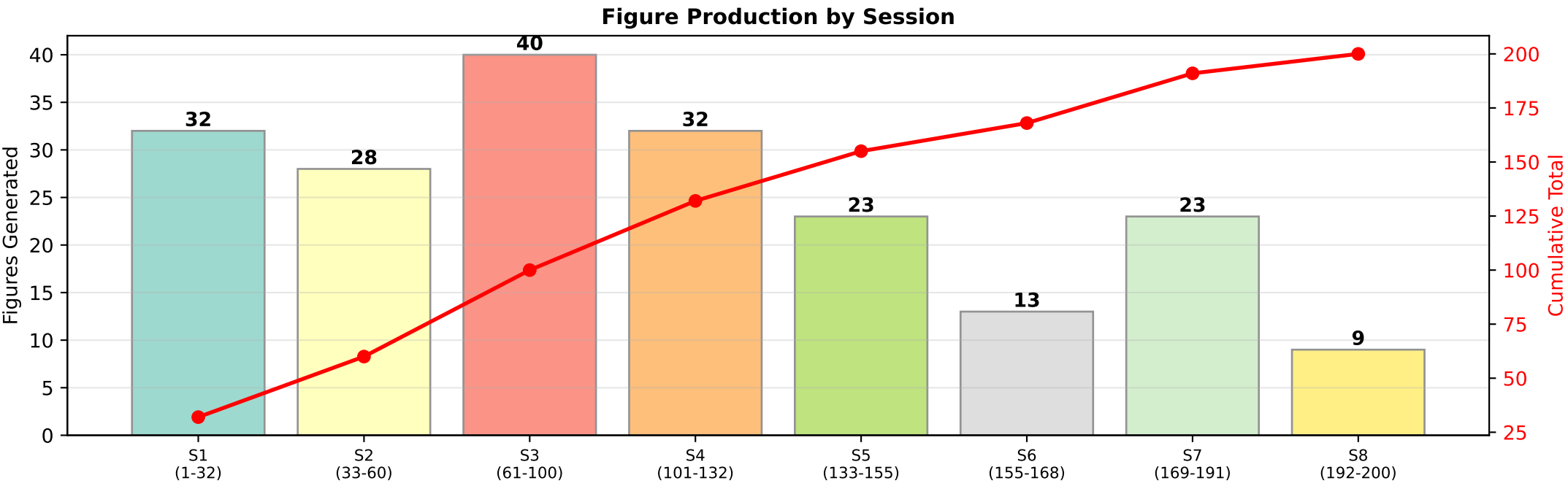
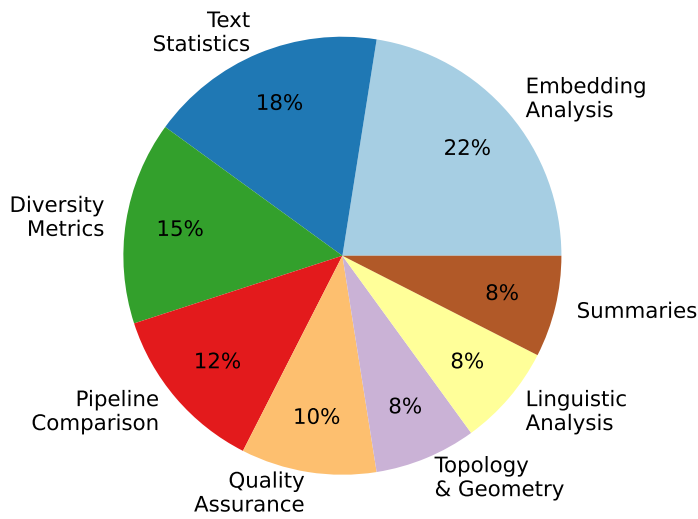
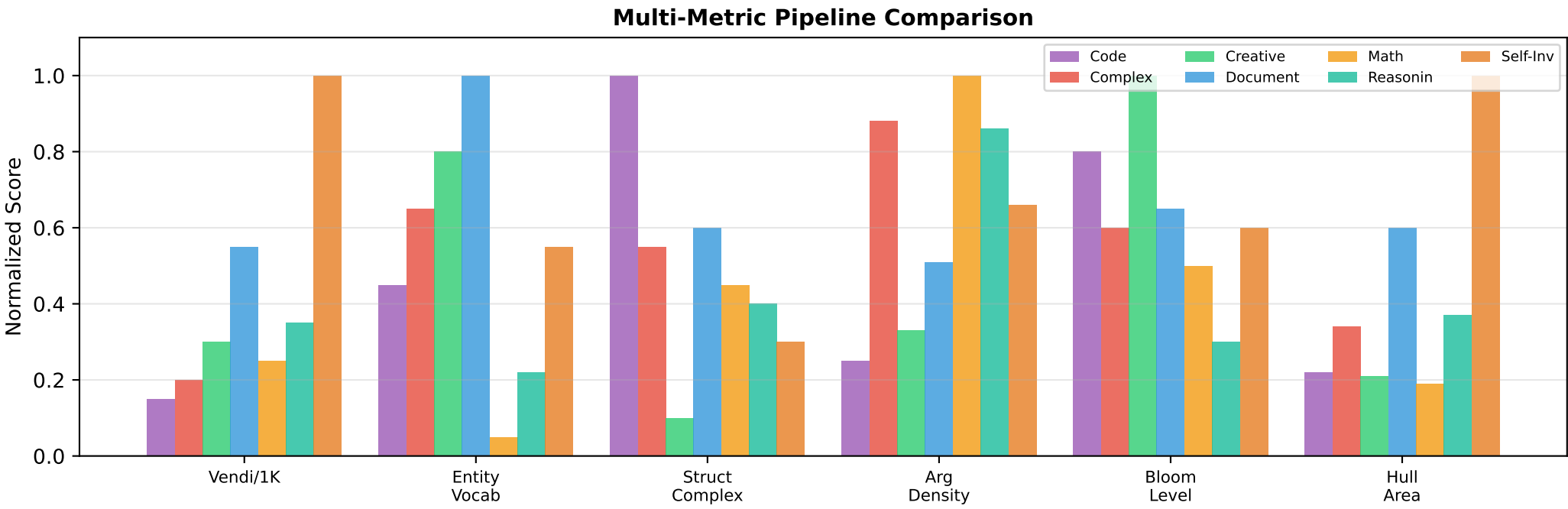


Figure Categories Distribution



Top 12 Quantitative Findings from 200 Figures	
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1.	Vendi Score 182.75 -- highest reported for synthetic data
2.	Self-Inversion: 2,288 unique domains (143x vs Complex QA)
3.	K-Center-Greedy: +33.3% diversity at N=500
4.	Cross-pipeline cosine distance: 0.961 (near-orthogonal)
5.	PCA requires 50+ dims for 90% variance (high intrinsic dim)
6.	Bootstrap rankings stable: CV < 5% across 30 resamples
7.	Document QA: L00 contribution +12.3 Vendi (largest impact)
8.	Greedy pipeline order: Self-Inv->DocQA->ComplexQA->...
9.	52,525 unique entities, 76.6% DocQA-exclusive
10.	Creative Writing: Bloom score 4.25 (highest cognitive)
11.	Code QA: 7.25 structural features/sample (most complex)
12.	Optimal mixing: upweight Self-Inv to 11% for +3% Vendi



Scaling & Efficiency Insights	
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Vendi Scaling: $V(N) = 55.4(1 - e^{(-N/745)}) + 109$ $R^2 = 0.996$, ceiling ~ 164.5	
Cost Efficiency:	
Self-Inversion:	\$12.3/1K (most efficient)
Document QA:	\$14.8/1K (Pareto-optimal)
Overall:	\$15.58/1K clean samples
Novelty Half-Life:	
Self-Inv:	0.9 batches (slowest decay)
Math:	0.3 batches (fastest saturation)
Tag Space: 4.1% utilized (96% unexplored)	

Project Status & Next Steps	
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COMPLETED:	
[x]	153,351 instruction samples
[x]	200 publication-quality figures
[x]	80+ JSON analysis reports
[x]	27 LaTeX tables
[x]	Complete paper draft (main.tex)
[x]	Training recipe (LLaMA-Factory)
[x]	Benchmark contamination check
IN PROGRESS:	
[]	Document QA: 21K/30K (token needed)
EXTENSIONS:	
-	Multi-lingual synthesis
-	Active learning selection
-	Human preference alignment